Project Report Format

Date	24th February 2025
Team ID	
Project Name	AI-Driven HealthCare Anomaly Detection
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INTRODUCTION

1.1 Project Overview: A real-time monitoring platform that identifies physiological anomalies in patient vitals using Deep Learning.

1.2 Purpose: To provide healthcare professionals with automated, early-warning alerts and AI-generated clinical insights to reduce alarm fatigue.

IDEATION PHASE

2.1 Problem Statement: Clinical staff are overwhelmed by high volumes of raw data and false alarms from traditional monitors.

2.2 Empathy Map Canvas: Focused on the emotional and cognitive load of nurses managing critical wards.

2.3 Brainstorming: Focused on integrating Kafka for streaming, TensorFlow for detection, and Gemini for insights.

REQUIREMENT ANALYSIS

3.1 Customer Journey Map: Mapping the transition from manual, reactive monitoring to proactive, AI-assisted care.

3.2 Solution Requirements: Defines functional needs like sub-second latency and persistent database logging.

3.3 Data Flow Diagram: Traces data from the Kafka Producer through AI inference to the Flask Dashboard.

3.4 Technology Stack: Utilizes Python, TensorFlow, Kafka, PostgreSQL, and Google Gemini API.

PROJECT DESIGN

4.1 Problem Solution Fit: Validating how the Autoencoder directly addresses the need for noise-reduction in clinical beeps.

4.2 Proposed Solution: An intelligent end-to-end pipeline that summarizes medical risks in plain language.

4.3 Solution Architecture: A multi-layered design ensuring scalability, reliability, and low-latency visualization.

PROJECT PLANNING & SCHEDULING

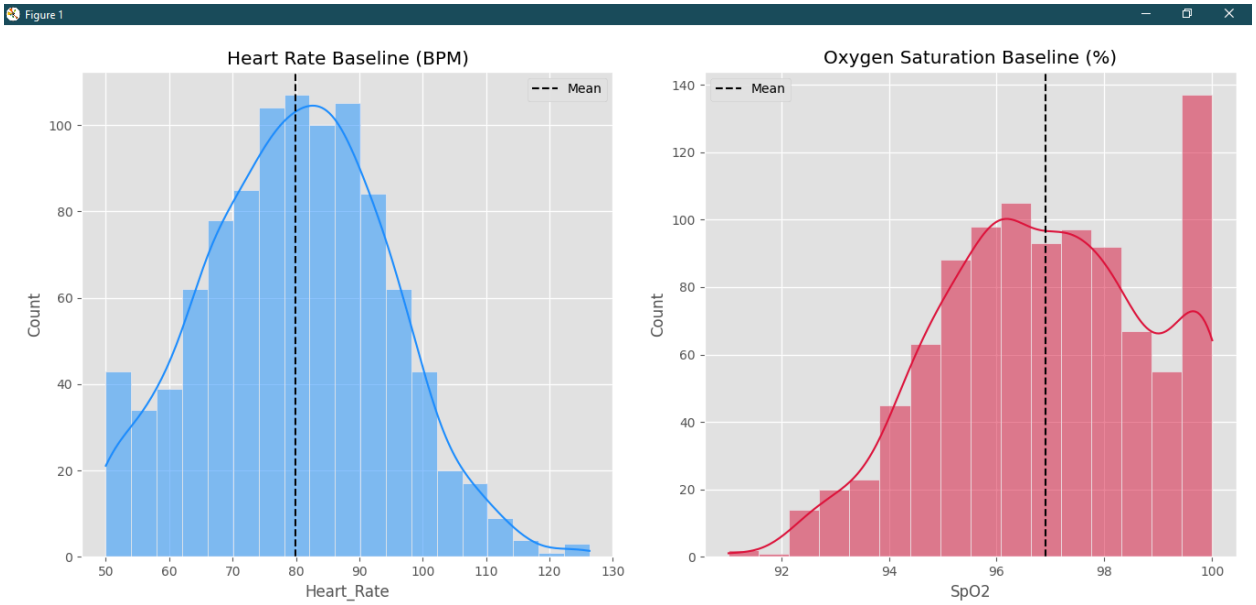
5.1 Project Planning: Managed through a 4-sprint Agile lifecycle with estimated story points and daily velocity tracking.

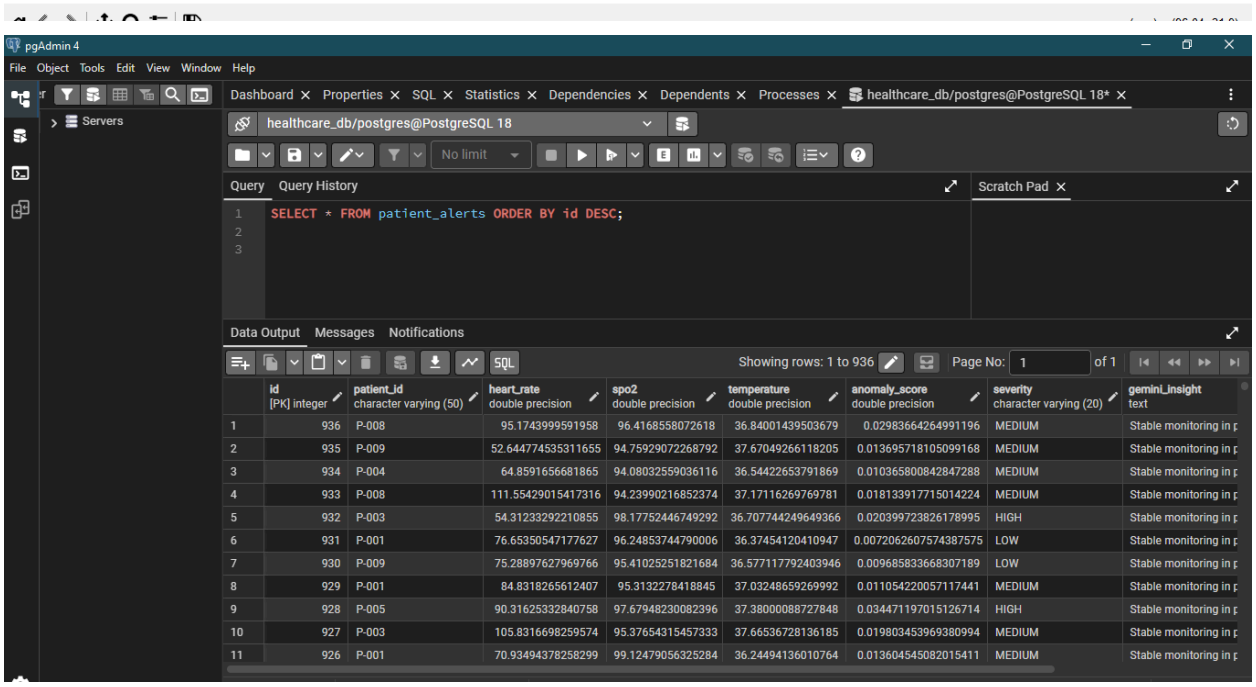
FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing: Verified end-to-end response times and successful anomaly flagging during simulated emergency events.

RESULTS

7.1 Output Screenshots: Validated through a live, responsive dashboard showing synchronized charts and clinical alert logs.





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2	935	P-009	52.644774535311655	94.75929072268792	37.67049266118205	0.013695718105099168	MEDIUM	Stable monitoring in p
3	934	P-004	64.8591656681865	94.08032559036116	36.54422653791869	0.010365800842847288	MEDIUM	Stable monitoring in p
4	933	P-008	111.55429015417316	94.23990216852374	37.17116269769781	0.018133917715014224	MEDIUM	Stable monitoring in p
5	932	P-003	54.31233292210855	98.17752446749292	36.707744249649366	0.020399723826178995	HIGH	Stable monitoring in p
6	931	P-001	76.65350547177627	96.24853744790006	36.37454120410947	0.0072062607574387575	LOW	Stable monitoring in p
7	930	P-009	75.28897627969766	95.41025251821684	36.577117792403946	0.009685833668307189	LOW	Stable monitoring in p
8	929	P-001	84.8318265612407	95.31322784188845	37.03248659269992	0.011054220057117441	MEDIUM	Stable monitoring in p
9	928	P-005	90.31625332840758	97.67948230082396	37.38000088727848	0.034471197015126714	HIGH	Stable monitoring in p
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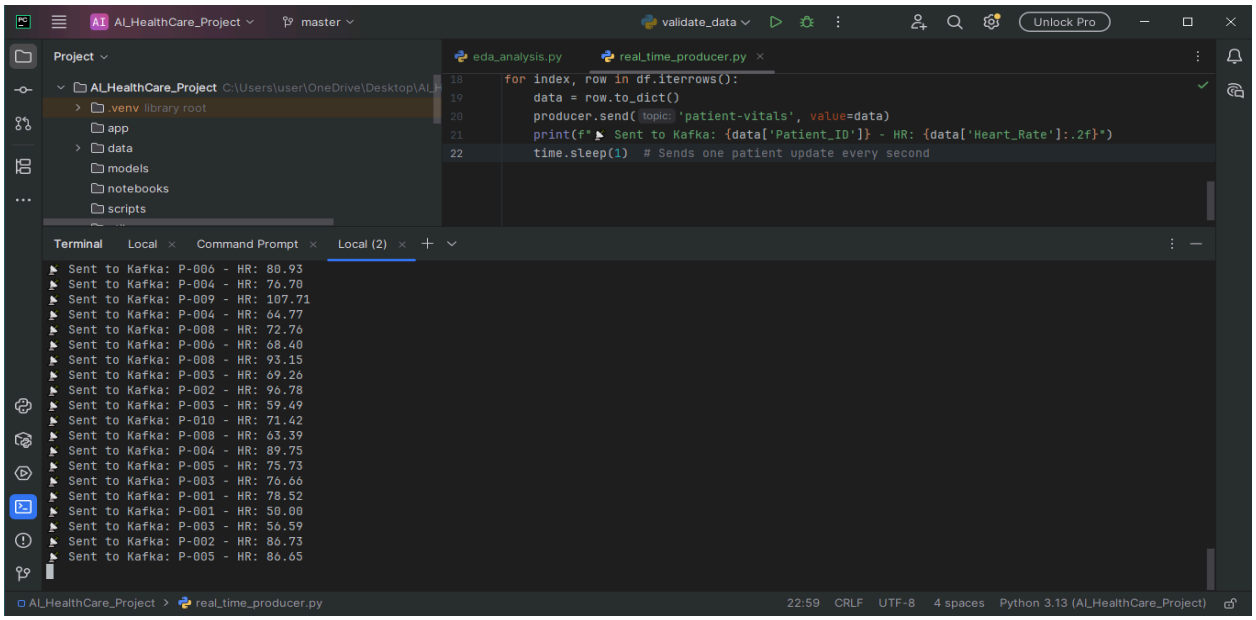
AI_HealthCare_Project
master
validate_data
Unlock Pro
Project
AI_HealthCare_Project C:\Users\user\OneDrive\Desktop\AI_H
.venv library root
app
data
models
notebooks
scripts
eda_analysis.py
real_time_producer.py
for index, row in df.iterrows():
data = row.to_dict()
producer.send(topic: 'patient-vitals', value=data)
print(f"📤 Sent to Kafka: {data['Patient_ID']} - HR: {data['Heart_Rate']:.2f}")
time.sleep(1) # Sends one patient update every second
Terminal
Local
Command Prompt
Local (2)
📤 Sent to Kafka: P-006 - HR: 80.93
📤 Sent to Kafka: P-004 - HR: 76.70
📤 Sent to Kafka: P-009 - HR: 107.71
📤 Sent to Kafka: P-004 - HR: 64.77
📤 Sent to Kafka: P-008 - HR: 72.76
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📤 Sent to Kafka: P-008 - HR: 93.15
📤 Sent to Kafka: P-003 - HR: 69.26
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📤 Sent to Kafka: P-003 - HR: 59.49
📤 Sent to Kafka: P-010 - HR: 71.42
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📤 Sent to Kafka: P-005 - HR: 75.73
📤 Sent to Kafka: P-003 - HR: 76.66
📤 Sent to Kafka: P-001 - HR: 78.52
📤 Sent to Kafka: P-001 - HR: 50.00
📤 Sent to Kafka: P-003 - HR: 56.59
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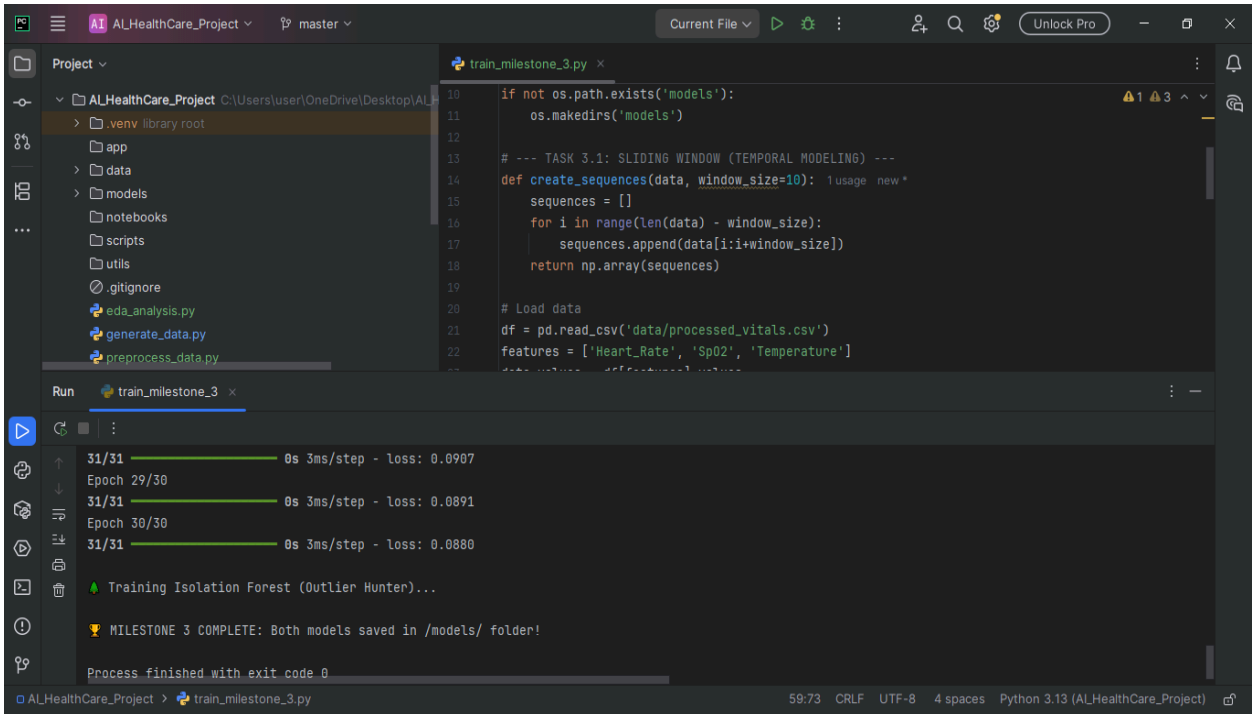
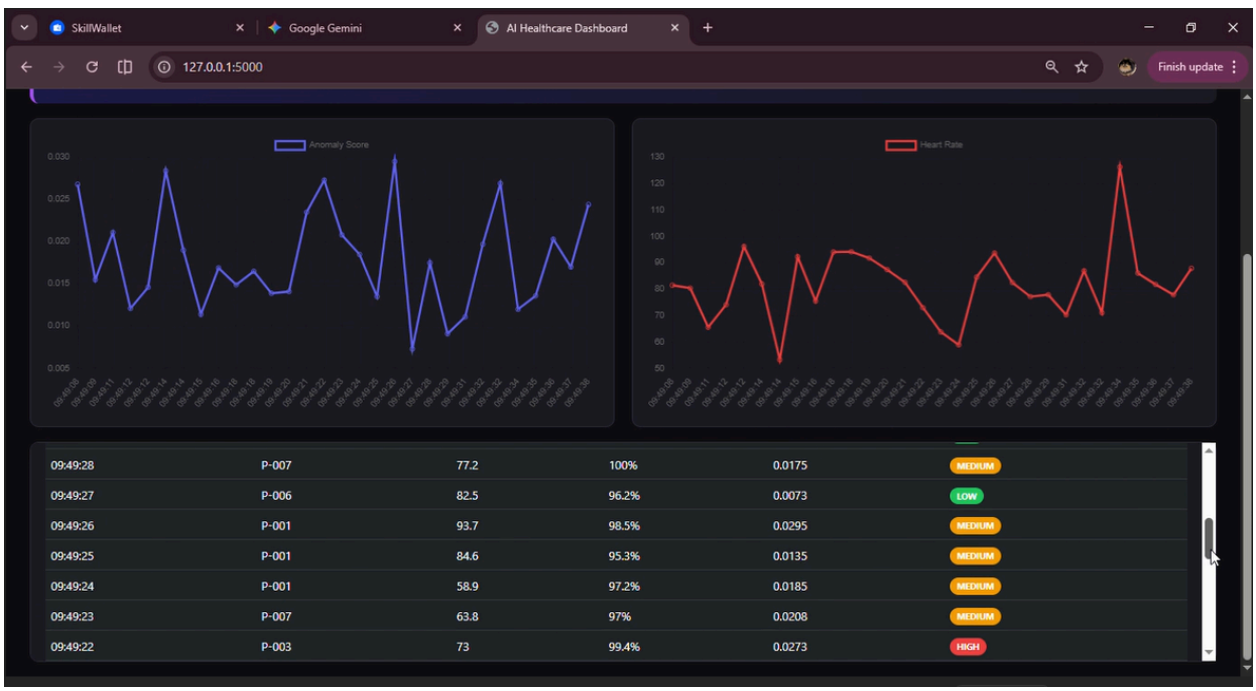
AI_HealthCare_Project > real_time_producer.py
22:59 CRLF UTF-8 4 spaces Python 3.13 (AI_HealthCare_Project)





ADVANTAGES & DISADVANTAGES

Advantages: Highly scalable, reduces nurse cognitive load, and provides historical audit trails.

Disadvantages: Requires reliable local infrastructure and active API connectivity for Generative AI features.

CONCLUSION

The system successfully bridges the gap between raw medical data and clinical action by providing "Predictive" detection and "Generative" context.

FUTURE SCOPE

Potential for expansion includes mobile app notifications and integration with physical wearable sensors.

APPENDIX

Source Code: Hosted on GitHub.

Dataset: 1,000 validated medical records.

GitHub Link:
https://github.com/kirti07k/AI-Driven-Healthcare-Anomaly-Detection-System